

In which quadrant are the following angles located?

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|-----------------------------|-------------------------|-------------------------------|-------------------------------|
| 1. $\theta = \frac{\pi}{3}$ | 3. $\theta = 3.5$ | 5. $\theta = \frac{15\pi}{4}$ | 7. $\theta = -\frac{5\pi}{3}$ |
| 2. $\theta = -115^\circ$ | 4. $\theta = 415^\circ$ | 6. $\theta = -550^\circ$ | 8. $\theta = -5.625$ |

Find a positive and a negative angle that is co-terminal with the given angle.

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|-----------------------|----------------------|-----------------------|-----------------------|
| 9. 438° | 11. $\frac{7\pi}{6}$ | 13. -105° | 15. $-\frac{3\pi}{4}$ |
| 10. $-\frac{9\pi}{2}$ | 12. 126° | 14. $\frac{7\pi}{12}$ | 16. -72° |

Find an angle that is a) complementary and b) supplementary to the given angle.
DO NOT CONVERT ANGLE MEASURES TO DEGREES!!!!

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|----------------------|-----------------------|----------------------|
| 17. $\frac{\pi}{6}$ | 19. $\frac{5\pi}{12}$ | 21. $\frac{5\pi}{3}$ |
| 18. $\frac{3\pi}{4}$ | 20. $\frac{9\pi}{4}$ | 22. $\frac{4\pi}{5}$ |

Convert the following angles from degrees to radians or from radians to degrees.

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|----------------------|----------------------|---------------------|-----------------------|
| 23. 135° | 25. $\frac{5\pi}{4}$ | 27. 240° | 29. $\frac{11\pi}{6}$ |
| 24. $\frac{2\pi}{3}$ | 26. 270° | 28. $\frac{\pi}{6}$ | 30. 330° |

Evaluate the following trig functions. NO CALCULATOR ALLOWED. (Hint: use the unit circle)

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|---------------------------|---------------------------|---------------------------|----------------------------|
| 33. $\sin \frac{2\pi}{3}$ | 35. $\cos \frac{3\pi}{4}$ | 37. $\cot \frac{5\pi}{4}$ | 39. $\csc \frac{11\pi}{6}$ |
| 34. $\tan \frac{\pi}{6}$ | 36. $\sec \pi$ | 38. $\cos \frac{5\pi}{3}$ | 40. $\sin \frac{3\pi}{2}$ |

Find the six trigonometric functions of θ (in standard position) whose terminal side passes through the given point. Give an exact answer. (Draw a picture)

41. (12, 16)

42. (3, -2)

Find the trigonometric functions of θ satisfying the condition. (Draw a picture)

43. Find $\sin \theta$ if $\sec \theta = \frac{6}{5}$ and $\tan \theta < 0$

47. Find $\sec \theta$ if $\tan \theta = 7$ and $\cos \theta < 0$

44. Find $\cos \theta$ if $\tan \theta = -\frac{12}{5}$ and $\sin \theta > 0$

48. Find $\cot \theta$ if $\sec \theta = \sqrt{7}$ and $\tan \theta < 0$

45. Find $\tan \theta$ if $\cos \theta = \frac{4}{7}$ and $\sin \theta > 0$

49. Find $\csc \theta$ if $\tan \theta = 5$ and $\sin \theta > 0$

46. Find $\cot \theta$ if $\csc \theta = \frac{3}{2}$ and $\cos \theta < 0$

50. Find $\cos \theta$ if $\sin \theta = \frac{2}{\sqrt{5}}$ and $\tan \theta < 0$

Find the exact values of θ . NO CALCULATOR & USE THE UNIT CIRCLE

51. $\sin \theta = \frac{1}{2}$

54. $\cot \theta = -\frac{\sqrt{3}}{3}$

57. $\tan \theta = \sqrt{3}$

60. $\cos \theta = -1$

52. $\cos \theta = -\frac{\sqrt{3}}{2}$

55. $\sec \theta = 2$

58. $\sin \theta = \frac{\sqrt{2}}{2}$

61. $\sin \theta = 0$

53. $\tan \theta = -1$

56. $\csc \theta = -2$

59. $\tan \theta = 0$

62. $\sec \theta = -\sqrt{2}$

Find the reference angle to the given angle. Answer in the same unit as the question. (Draw a picture)

63. $\theta = 7\pi/6$

65. $\theta = 87^\circ$

67. $\theta = 2\pi/3$

64. $\theta = -125^\circ$

66. $\theta = 3.56$

68. $\theta = 2.5$

Use a calculator on # 69 – 82.

Evaluate the following to the nearest ten-thousandth (4 d.p.). (Make sure calculator is in degrees or radians depending on the question!)

69. $\sin 15^\circ$

72. $\cos 495^\circ$

70. $\sec 2.45$

73. $\tan -155^\circ$

71. $\cot 1.5$

74. $\csc -3.2$

Find the measures of θ so that $0^\circ \leq \theta \leq 360^\circ$ to 2 decimal places. (You should have more than one answer for each question)

75. $\sin \theta = 0.2586$

79. $\csc \theta = 1.5$

76. $\cos \theta = -0.2679$

80. $\sin \theta = -0.2387$

77. $\tan \theta = 6.75$

81. $\cos \theta = 0.0042$

78. $\cot \theta = -3/2$

82. $\sec \theta = 154/17$

On #83-91, give exact answers (no decimals). Be sure to include units in your answer.

83. Find the radian measure of the central angle of a circle of radius of 11 feet that intercepts an arc of length 5 feet.

84. Find the radian measure of the central angle of a circle of radius of 12.5 feet that intercepts an arc of length 25 feet.

85. Find the length of the arc on a circle of radius 27 meters intercepted by a central angle $\frac{2\pi}{3}$ radians.

86. Find the length of the arc on a circle of radius 9 feet intercepted by a central angle of 60° .

87. Find the radius of a circle with the arc length 41 feet and central angle 135° .

88. Find the radius of a circle with the arc length 64 feet and central angle 210° .

89. When a laser reads a compact disc at the outer edge, it travels at a speed of 200 rpm.

a. Through how many radians does it travel in a minute?

b. If the diameter of the disc is 11.9 cm, find the distance that a piece of dust on the outer edge of the CD travels in 1 minute.

c. What is the linear speed in cm/second?

90. A potter's wheel turns at a rate of 120 rpm. If the diameter of the wheel is 13 inches find the distance traveled by a point on the outer edge in one minute.

91. A wheel with radius of 2 in takes 3 seconds to make 6 revolutions. What is the angular speed of the wheel? What is the linear speed?

92. A bicycle has a tire radius of 12 inches. It takes 2 seconds to rotate 10 revolutions. How fast is the wheel turning (angular speed)? How fast is the bicycle travelling in miles per hour (linear speed)? You may use a calculator to determine the exact and approximate linear speed in miles per hour.

Review all other application problems assigned.